

Python Lab Task

Setting up VS Code, Git & GitHub — Version Control Basics — Submitting a Python Assignment via Git

1. Installing VS Code, Git and GitHub

Step 1: Install Visual Studio Code

Visit the official website: <https://code.visualstudio.com/> and download the installer for your operating system (Windows / macOS / Linux). Run the installer and accept the default options. On Windows, make sure to check the box **'Add to PATH'** during installation.

Step 2: Install Git

Visit <https://git-scm.com/downloads> and download Git for your operating system. During installation, keep the default settings (default editor, PATH environment variable, etc.). After installation, verify it by opening a terminal and running:

```
git --version
```

Step 3: Create a GitHub Account

Go to <https://github.com/> and sign up for a free account (skip this step if you already have one).

Step 4: Configure Git Identity

Before using Git for the first time, set your username and email (this information is attached to every commit you make):

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

2. Essential Git Commands

Command	Purpose
git init	Initialize a new Git repository in the current folder
git status	Check status of tracked / untracked / modified files
git add <file>	Stage a specific file for commit
git add .	Stage all changed files in the folder
git commit -m "message"	Commit staged changes with a message
git log	View commit history
git branch	List all branches
git checkout -b <name>	Create and switch to a new branch
git remote add origin <url>	Link the local repository to a GitHub repo

Command	Purpose
<code>git push -u origin main</code>	Push commits to GitHub (first time)
<code>git push</code>	Push commits to GitHub (subsequent times)
<code>git pull</code>	Fetch and merge changes from GitHub
<code>git clone <url></code>	Copy a GitHub repository to your local machine

3. Connecting Git and GitHub in VS Code

Install Required Extensions

Open VS Code → go to the Extensions panel (Ctrl+Shift+X) → install the following:

- **GitHub Pull Requests and Issues** (by GitHub) — lets you sign in to GitHub directly and manage pull requests / issues from within VS Code.
- **GitLens** (optional but recommended) — adds detailed commit history, blame annotations, and repository insights.
- Git itself needs no separate extension — VS Code has built-in Git support available in the **Source Control** tab.

Sign in to GitHub from VS Code

- 1 Click the Accounts icon in the bottom-left corner of VS Code.
- 2 Select **'Sign in with GitHub'**.
- 3 A browser window will open — authorize VS Code to access your account.
- 4 You are now connected and ready to push / pull repositories.

Submitting an Assignment via VS Code (Step-by-Step)

- 1 Create a folder for your assignment and open it in VS Code (*File > Open Folder*).
- 2 Open the integrated terminal using **Ctrl + `**.
- 3 Initialize a Git repository: `git init`
- 4 Create your Python file (see Section 4 below).
- 5 Go to the **Source Control** tab (branch-shaped icon in the left sidebar).
- 6 Stage your file by clicking the '+' icon next to it, or run `git add .`
- 7 Type a commit message and click the checkmark button (or run `git commit -m "message"`).
- 8 Create a new repository on GitHub (via the browser), or use VS Code's **'Publish to GitHub'** button directly from the Source Control panel.
- 9 Push your code: `git push -u origin main`

Note: Once pushed, share your GitHub repository link with your instructor as proof of submission.

4. Simple Python Program & Push to GitHub

Sample Program — hello.py

```
def greet(name):  
    return f"Hello, {name}! Welcome to Git and GitHub."  
  
if __name__ == "__main__":  
    user_name = input("Enter your name: ")  
    print(greet(user_name))
```

Commands to Push This Program to GitHub

```
# 1. Navigate to your project folder  
cd path/to/your/folder  
  
# 2. Initialize git repo  
git init  
  
# 3. Stage the file  
git add hello.py  
  
# 4. Commit  
git commit -m "Add simple python greeting program"  
  
# 5. Link to your GitHub repo (create the repo on github.com first)  
git remote add origin https://github.com/<your-username>/<repo-name>.git  
  
# 6. Rename branch to main (if needed)  
git branch -M main  
  
# 7. Push to GitHub  
git push -u origin main
```

After running these commands, refresh your GitHub repository page in the browser — **hello.py** should now appear there. This confirms your assignment has been submitted successfully.